Veterinary Oncology Drugs — NAVLE Cheat Sheet

Drug | class & mechanism | what it treats | what it does to the patient | species traps. Bold = the detail most likely to be the answer.

THE SIX FACTS THAT SHOW UP MOST

1. **Cyclophosphamide** → **sterile hemorrhagic cystitis** (acrolein). Prevent with AM dosing + water ± furosemide; treat with mesna.
2. **Doxorubicin** → **dilated cardiomyopathy in DOGS** (cumulative, dose-dependent) but **nephrotoxicity in CATS**. Severe vesicant.
3. **Cisplatin is lethal in cats** — fatal pulmonary edema. Use carboplatin instead. Cisplatin in dogs = nephrotoxic, needs saline diuresis.
4. **5-FU is lethal in cats** — seizures and death, including from licking topical cream off an owner.
5. **Vincristine** → **peripheral neuropathy + ileus; vinblastine** → **myelosuppression.** Vincristine is the drug of choice for TVT.
6. **Lomustine** → **hepatotoxicity + delayed cumulative thrombocytopenia**; crosses the BBB.

Alkylating agents

Drug	Class / MOA	Main uses	Key toxicities	Species / high-yield
Cyclo-phosphamide	Alkylating agent (nitrogen mustard) — cross-links DNA; cell-cycle nonspecific	Lymphoma (the **C** in CHOP); sarcomas, carcinomas; immune-mediated disease (IMHA, ITP)	**Sterile hemorrhagic cystitis** from the acrolein metabolite; myelosuppression (nadir ~7–10 d); alopecia (poodles, OES, terriers); GI upset	Cystitis is the classic board answer. Prevent: dose in AM, free water, ± furosemide. Treat/prevent with **mesna**. Do <u>not</u> re-challenge after cystitis.
Chlorambucil	Alkylating agent — slowest, gentlest of the class; oral	Feline **small-cell (low-grade) GI lymphoma**; chronic lymphocytic leukemia; IBD; immune-mediated disease	Gradual, cumulative myelosuppression (thrombocytopenia); GI upset; rare reversible **myoclonus/neurotoxicity in cats** at high dose	Workhorse oral drug in cats. Well tolerated; monitor CBC q2–4 wk.
Lomustine (CCNU)	Nitrosourea alkylator — **lipid soluble, crosses the blood–brain barrier**; oral	Mast cell tumor; rescue lymphoma; **CNS lymphoma**; cutaneous epitheliotropic (T-cell) lymphoma; histiocytic sarcoma	**Hepatotoxicity in dogs** (cumulative, can be fatal — monitor ALT); neutropenia nadir ~7 d; **delayed cumulative thrombocytopenia**; pulmonary fibrosis in cats	Two-toxicity drug: liver + delayed marrow. Give SAMe/silybin and check ALT before every dose.
Melphalan	Alkylating agent (phenylalanine mustard); oral	**Multiple myeloma** (with prednisone) — the standard answer	Myelosuppression, esp. thrombocytopenia; cumulative	Myeloma = melphalan + prednisone. Monitor CBC.
Ifosfamide	Alkylating agent, cyclophosphamide analog	Soft tissue sarcoma; rescue protocols	**Severe hemorrhagic cystitis** and nephrotoxicity — mandatory mesna + saline diuresis; myelosuppression	Never give without mesna and diuresis.
Dacarbazine (DTIC)	Alkylating/antimetabolite hybrid	Rescue lymphoma (e.g., with doxorubicin); sarcomas	Severe vomiting; myelosuppression; **vesicant** — perivascular necrosis	Notorious emetic — pre-treat with maropitant.

Antitumor antibiotics

Drug	Class / MOA	Main uses	Key toxicities	Species / high-yield
Doxorubicin	Anthracycline antitumor antibiotic — topoisomerase II inhibitor + free-radical DNA damage	Lymphoma (the **H** in CHOP); **hemangiosarcoma**; osteosarcoma; carcinomas; single-agent rescue	**Dogs: cumulative, dose-dependent dilated cardiomyopathy** (cap ~180–240 mg/m² cumulative); **severe vesicant**; acute anaphylactoid histamine release during infusion; myelosuppression nadir 7–10 d; hemorrhagic colitis 3–5 d; red-orange urine	**Cats get nephrotoxicity, not cardiotoxicity.** Screen dogs with echo; avoid in pre-existing DCM/Dobermans/Boxers. **Dexrazoxane** = cardioprotectant + extravasation antidote. Slow infusion + antihistamine.
Mitoxantrone	Anthracenedione — topo II inhibitor; less cardiotoxic than doxorubicin	Transitional cell carcinoma (with piroxicam); lymphoma rescue; SCC in cats	Myelosuppression (dose-limiting); GI upset; **blue-green urine/sclera**	Substitute when doxorubicin cardiotoxicity is a concern.
Actinomycin D (dactinomycin)	Antitumor antibiotic — intercalates DNA	Doxorubicin substitute in lymphoma protocols	Myelosuppression; GI; vesicant	Used when the doxorubicin cumulative cardiac dose is reached.

Antimetabolites

Drug	Class / MOA	Main uses	Key toxicities	Species / high-yield
Cytarabine (Ara-C)	Pyrimidine antimetabolite — **S-phase specific**; crosses the BBB	**CNS lymphoma**; acute leukemias; also immunosuppressant for MUE/GME	Myelosuppression; GI upset	The 'gets into the CNS' antimetabolite.
Methotrexate	Folate antagonist — inhibits dihydrofolate reductase	Older lymphoma protocols (e.g., COAP/L-MOPP era)	GI ulceration/mucositis; myelosuppression; hepatotoxicity	**Rescue = leucovorin (folinic acid).** Renal excretion — avoid with NSAIDs.
5-Fluorouracil (5-FU)	Pyrimidine antimetabolite	Dogs: carcinomas; topical for cutaneous tumors	**Fatal neurotoxicity — seizures, cerebellar signs**; GI; myelosuppression	**ABSOLUTELY CONTRAINDICATED IN CATS.** Also a common toxicosis question: cat licks owner's topical 5-FU cream → seizures → death.
Gemcitabine	Pyrimidine antimetabolite; **radiosensitizer**	Carcinomas; combined with radiation or carboplatin	Myelosuppression; GI	Think 'radiation sensitizer.'

Mitotic spindle inhibitors (vinca alkaloids & taxanes)

Drug	Class / MOA	Main uses	Key toxicities	Species / high-yield
Vincristine	Vinca alkaloid — binds tubulin, blocks mitotic spindle (**M-phase**)	Lymphoma (the **O**/Oncovin in CHOP); **transmissible venereal tumor (TVT) — drug of choice**; ITP (raises platelet count)	**Peripheral neuropathy**; ileus/constipation; **severe vesicant**; relatively mild myelosuppression	Neurotoxic > myelosuppressive. TVT and ITP are the two signature non-lymphoma uses.
Vinblastine	Vinca alkaloid — same MOA as vincristine	**Mast cell tumor** (with prednisone ± lomustine)	**Myelosuppression is dose-limiting** (nadir ~7 d); vesicant; less neurotoxic than vincristine	Flip of vincristine: more marrow, less nerve.
Paclitaxel / docetaxel	Taxane — stabilizes microtubules, blocks mitosis	Carcinomas, mast cell tumor (limited use)	Hypersensitivity to the vehicle (Cremophor) in dogs; myelosuppression	Pre-medicate for hypersensitivity.

Platinum agents

Drug	Class / MOA	Main uses	Key toxicities	Species / high-yield
Cisplatin	Platinum agent — DNA cross-links	Canine **osteosarcoma** (post-amputation); carcinomas; intracavitary for mesothelioma	**Nephrotoxicity** (requires aggressive saline diuresis); severe emesis; ototoxicity; myelosuppression	**NEVER IN CATS — fatal pulmonary edema.** This is one of the most heavily tested facts on the exam.
Carboplatin	Platinum agent — same MOA, better tolerated	Osteosarcoma (dogs); carcinomas; **the platinum you can use in cats** (e.g., injection-site sarcoma, SCC)	**Myelosuppression is dose-limiting**; thrombocytopenia with delayed nadir in cats; minimal nephro-/oto-toxicity; less emetic	Safe in cats, no diuresis needed. Dose off GFR/body weight in small dogs.

Enzymes, hormones, targeted agents & other

Drug	Class / MOA	Main uses	Key toxicities	Species / high-yield
L-asparaginase	Enzyme — depletes extracellular asparagine, which lymphoblasts cannot synthesize	Lymphoma/lymphoid leukemia induction and rescue; fast, marrow-sparing response	**Anaphylaxis** (give SC or IM, <u>not</u> IV); **pancreatitis**; coagulopathy; little myelosuppression	Do not give on the same day as vincristine (↑ neutropenia and pancreatitis risk) — separate by 6+ h or dose the day before.
Prednisone / prednisolone	Glucocorticoid — induces apoptosis in lymphoid cells	Lymphoma (the **P** in CHOP); mast cell tumor; brain tumor edema; palliative care; hypercalcemia of malignancy	PU/PD/PP, iatrogenic hyperadrenocorticism, GI ulceration (worse with NSAIDs), immunosuppression, muscle wasting	**Never start steroids before you have a diagnosis** — they destroy cytology/histopath, mask disease, and induce multidrug resistance (shorter remission).

Drug	Class / MOA	Main uses	Key toxicities	Species / high-yield
Hydroxyurea	Ribonucleotide reductase inhibitor — S-phase	**Polycythemia vera**; chronic myelogenous leukemia; mast cell tumor	Myelosuppression; **methemoglobinemia and anemia in cats**; onychomadesis (nail sloughing) in dogs	The go-to for polycythemia vera after phlebotomy.
Toceranib (Palladia)	Receptor tyrosine kinase inhibitor — **KIT**, VEGFR, PDGFR; oral	Labeled for recurrent grade II/III **cutaneous mast cell tumor**; off-label for anal sac adenocarcinoma, thyroid carcinoma, others	**Diarrhea/hemorrhagic GI signs** (dose-limiting); anorexia; neutropenia; **hypertension and proteinuria**; hypoalbuminemia; muscle cramping; lameness	Works best on tumors with an activating **c-KIT** mutation. Monitor BP and urine protein:creatinine.
Masitinib (Kinavet)	Tyrosine kinase inhibitor — KIT/PDGFR	Canine mast cell tumor	**Protein-losing nephropathy**; GI; neutropenia	Check UPC before and during therapy.
Piroxicam	Non-selective NSAID — COX inhibition, antiangiogenic	**Transitional cell carcinoma of the bladder** (± mitoxantrone); SCC; other carcinomas	GI ulceration/perforation; **renal papillary necrosis**; hepatopathy	TCC = piroxicam. Give with misoprostol/gastroprotectant; never combine with corticosteroids.
Rabacfosadine (Tanovea)	Acyclic nucleotide prodrug targeted to lymphoid cells	Canine **lymphoma** (conditionally approved, single agent)	**Pulmonary fibrosis** (can be fatal); dermatopathy; GI; myelosuppression	Avoid in West Highland White Terriers and other breeds predisposed to pulmonary fibrosis.
Tigilanol tiglate (Stelfonta)	Intratumoral protein kinase C activator — destroys tumor vasculature	Non-metastatic canine **mast cell tumor** (single intratumoral injection)	Massive **degranulation** → hypotension, GI ulceration; large open wound at the site that heals by second intention	Mandatory pre-treatment: H1 blocker + H2 blocker + corticosteroid.

Quick-reference tables

Buzzword → drug	
Sterile hemorrhagic cystitis	Cyclophosphamide (or ifosfamide)
Dilated cardiomyopathy in a dog	Doxorubicin
Nephrotoxicity in a cat on chemo	Doxorubicin
Fatal pulmonary edema in a cat	Cisplatin
Seizures in a cat after topical cream	5-Fluorouracil
Peripheral neuropathy / ileus	Vincristine
Blue-green urine and sclera	Mitoxantrone
Red-orange urine	Doxorubicin
Hepatotoxicity + delayed thrombocytopenia	Lomustine
Anaphylaxis + pancreatitis	L-asparaginase
Renal papillary necrosis	Piroxicam (NSAID)
Proteinuria + hypertension + diarrhea	Toceranib
Pulmonary fibrosis in a Westie	Rabacfosadine
Myoclonus in a cat	Chlorambucil
Methemoglobinemia in a cat	Hydroxyurea

Tumor → first-line drug	
Multicentric lymphoma (dog)	CHOP protocol
Feline small-cell GI lymphoma	Chlorambucil + prednisolone
CNS lymphoma	Lomustine or cytarabine (cross BBB)
Transmissible venereal tumor	Vincristine
Mast cell tumor	Vinblastine + pred, lomustine, toceranib
Osteosarcoma (post-amputation)	Carboplatin (or cisplatin, dogs)
Hemangiosarcoma (post-splenectomy)	Doxorubicin
Bladder transitional cell carcinoma	Piroxicam ± mitoxantrone
Multiple myeloma	Melphalan + prednisone
Polycythemia vera	Hydroxyurea
Anal sac adenocarcinoma	Toceranib / carboplatin
Injection-site sarcoma (cat)	Surgery + RT ± doxorubicin/carboplatin

Toxicity → antidote / management	
Doxorubicin extravasation	Stop, aspirate, cold compress, **dexrazoxane** IV
Vincristine/vinblastine extravasation	Warm compress + hyaluronidase (do NOT cool)
Cyclophosphamide cystitis	Discontinue permanently; mesna, furosemide, fluids
Methotrexate overdose	Leucovorin (folinic acid)
Cisplatin nephrotoxicity	Saline diuresis before/after; never use in cats
Anthracycline cardiotoxicity	Prevent: echo screening, cumulative dose cap, dexrazoxane
Febrile neutropenia	Broad-spectrum antibiotics, fluids, hospitalize; dose-reduce 20–25%
Chemo-induced nausea/vomiting	Maropitant, ondansetron; metoclopramide
Mast cell degranulation	H1 + H2 blockers ± steroid; pre-treat before cytoreduction

SPECIES & BREED CAUTIONS

Cats: no cisplatin (fatal pulmonary edema), no 5-FU (fatal neurotoxicity). Doxorubicin causes **nephrotoxicity**, not cardiotoxicity — check renal values. Lomustine can cause pulmonary fibrosis. Chlorambucil and carboplatin are the well-tolerated staples. Use prednisoLONE, not prednisone.

Dogs with the MDR1 (ABCB1-1Δ) mutation — Collies, Australian Shepherds, Shelties and other herding breeds: reduced clearance of vincristine, vinblastine, doxorubicin and actinomycin D → severe myelosuppression. Test and dose-reduce.

Doberman, Boxer, Great Dane: pre-existing DCM risk — echo before doxorubicin. **West Highland White Terrier:** avoid rabacfosadine. **Greyhounds/sighthounds:** low body fat, higher toxicity risk.

PROTOCOL & SAFETY POINTS WORTH A FREE POINT

CHOP = **C**yclophosphamide, **H**ydroxydaunorubicin (doxorubicin), **O**ncovin (vincristine), **P**rednisone — 19–25 wk, the standard for canine multicentric lymphoma. Median survival ~12 mo; ~80–90% remission. Prednisone alone: ~2–3 mo.

Check a CBC before every dose. Typical neutrophil nadir is 7 days (5–10 d). Delay treatment if neutrophils are below ~1,500–2,000/µL; reduce the next dose by 20–25% after grade 3–4 neutropenia.

Substage b (clinically ill) and **T-cell immunophenotype** = worse prognosis. Hypercalcemia in a dog with lymphoma suggests **mediastinal/T-cell** disease.

Handling: chemotherapy agents are hazardous drugs — chemo gloves, closed-system transfer devices, no compounding by pregnant staff, and owners must handle patient waste with gloves for 3–5 days after treatment.

Cell-cycle specific (vincristine = M, cytarabine/hydroxyurea = S) vs **nonspecific** (alkylators, doxorubicin) — specific drugs favor frequent low doses; nonspecific favor higher intermittent dosing.

Study aid only — verify all doses and protocols against a current formulary before clinical use. Prepared July 2026.